

Quantity Theory, Inflation, and the Demand for Money

Chapter 19

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

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Preview and Learning Objectives

This chapter examines the quantity theory of money and its link to the demand for money.

Key question: how much money do people want to hold, and what does that imply for the price level and inflation?

Outline

Quantity Theory of Money

Budget Deficits and Inflation

Liquidity Preference Theory

Portfolio Choice Theory

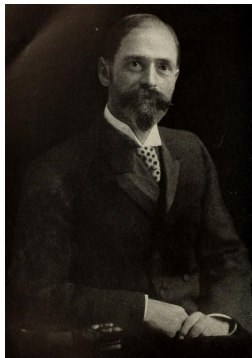
Empirical Evidence on Money Demand

Quantity Theory of Money: Equation of Exchange

Equation of Exchange

$$M \times V = P \times Y$$

- ▶ M : money supply
- ▶ V : velocity of money (how frequently money is used in transactions)
- ▶ P : price level
- ▶ Y : real output (income)
- ▶ PY : nominal GDP



Irving Fisher (1867–1947), whose equation of exchange underlies the quantity theory.

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Quantity Theory of Money: Assumptions

Assumptions

- ▶ Velocity (V) is relatively **constant**.
- ▶ Real output (Y) is determined by real factors and remains at the **full-employment level**.

Fisher's Interpretation

If velocity is constant,

$$P \times Y = M \times \bar{V}$$

Nominal income is determined by the **quantity of money**.

Implication

Changes in the money supply affect mainly the **price level**.

Quantity Theory: Money Demand and the Price Level

Money Demand

Rewrite the equation with $k = 1/V$:

$$M = \frac{1}{V} \times PY$$

$$M^d = k \times PY$$

Price Level Determination

Divide both sides by $\bar{Y}$:

$$P = \frac{M \times \bar{V}}{\bar{Y}}$$

Implication

The price level depends on the money supply M , velocity V , and real output Y .

Quantity Theory and Inflation

Growth Rate Form

Taking percentage changes:

$$\% \Delta M + \% \Delta V = \% \Delta P + \% \Delta Y$$

Inflation Equation

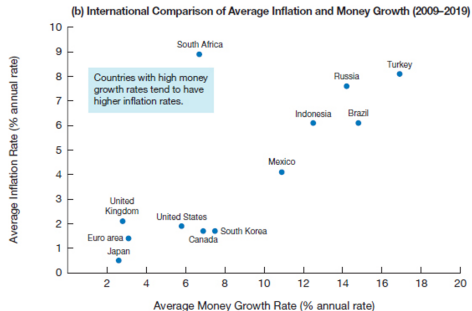
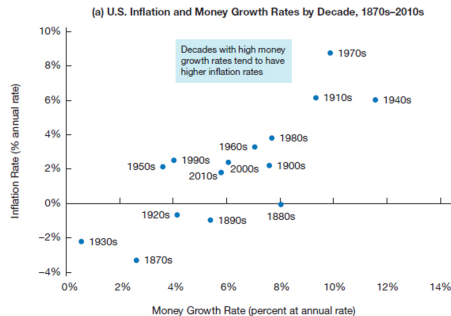
$$\pi = \% \Delta M + \% \Delta V - \% \Delta Y$$

If velocity is constant:

$$\pi = \% \Delta M - \% \Delta Y$$

Inflation is the money growth rate minus the growth rate of real output: “too much money chasing too few goods.”

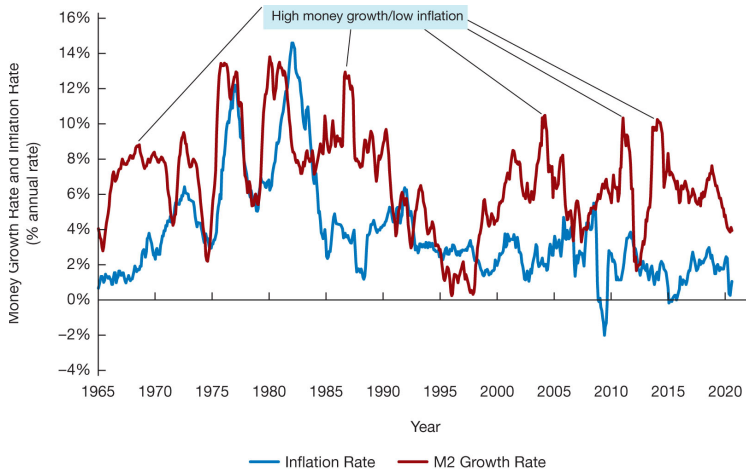
Relationship Between Inflation and Money Growth



Average inflation and money growth across countries.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

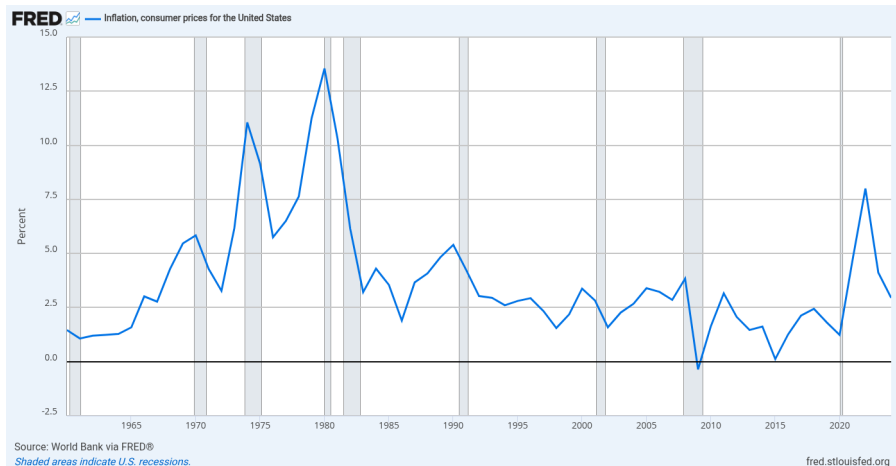
U.S. Inflation and Money Growth, 1965–2019



Annual U.S. inflation and M2 growth rates.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

U.S. CPI Inflation Since 1960



Annual U.S. consumer price inflation, percent; note the 1970s peaks and the 2021–22 episode.

Source: FRED, Federal Reserve Bank of St. Louis

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Budget Deficits and Inflation

Government Budget Constraint

$$DEF = G - T = \Delta MB + \Delta B$$

- ▶ DEF = budget deficit (government spending G minus taxes T)
- ▶ ΔMB = increase in the **monetary base** (money printing)
- ▶ ΔB = increase in **government bonds** held by the public

Financing Government Deficits

Two Ways to Finance a Deficit

- ▶ **Borrowing from the public** (ΔB)
 - ▶ Government sells bonds
 - ▶ No direct effect on the money supply
- ▶ **Printing money** (ΔMB)
 - ▶ Central bank creates money to buy government debt
 - ▶ Increases the money supply
 - ▶ May lead to inflation

Implication

Not all budget deficits cause inflation. Inflation occurs when deficits are financed by **printing money** rather than borrowing.

Deficits Financed by Printing Money: Hyperinflations



German 10 billion ("Zehn Milliarden") mark Reichsbank note, 1923.

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Zimbabwe 100 trillion dollar note, 2009.

*Wikimedia Commons, Reserve Bank of Zimbabwe,
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When a government covers its deficit by having the central bank create money (ΔMB), the money supply grows and so does the price level.

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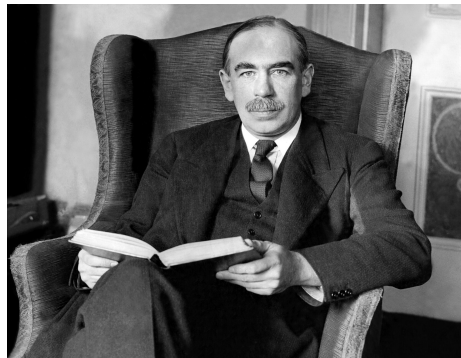
Empirical Evidence on Money Demand

Keynesian Theories of Money Demand

Liquidity Preference Theory

Keynes proposed that individuals hold money for three motives:

- ▶ **Transactions motive**
- ▶ **Precautionary motive**
- ▶ **Speculative motive**



John Maynard Keynes (1883–1946), 1929.

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Transactions and Precautionary Motives

Transactions motive

- ▶ People hold money to carry out everyday transactions.
- ▶ Higher income $\Rightarrow$ more transactions $\Rightarrow$ higher demand for money.

Precautionary motive

- ▶ People hold money as a cushion against unexpected expenses (medical emergencies, unexpected bills).
- ▶ Precautionary balances also increase with income.

$$M^d \propto Y$$

Speculative Motive and Real Money Demand

- ▶ Money can be held as a store of wealth.
- ▶ Individuals choose between holding money and holding bonds.
- ▶ The opportunity cost of holding money is the **interest rate**: $M^d \downarrow$ when $i \uparrow$.

Putting the Three Motives Together: Real Money Demand

$$\frac{M^d}{P} = f(i, Y)$$

- ▶ Negatively related to interest rates i
- ▶ Positively related to real income Y

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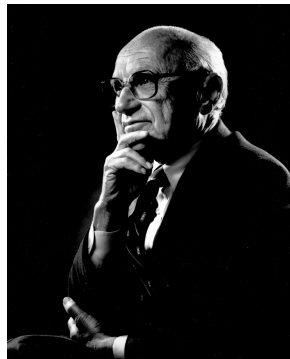
Portfolio Choice Theory

Empirical Evidence on Money Demand

Portfolio Theories of Money Demand

- ▶ Individuals allocate wealth across different assets.
- ▶ The demand for money depends on several factors:
 - ▶ Wealth
 - ▶ Risk of other assets
 - ▶ Liquidity of other assets

Money is one asset in a portfolio: its demand rises with wealth and with the risk of other assets, and falls with the liquidity of other assets.



Milton Friedman (1912–2006), who treated money demand as a portfolio choice.

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Factors That Determine the Demand for Money

Variable	Change in Variable	Money Demand Response	Reason
Interest rates	↑	↓	Opportunity cost of money rises
Income	↑	↑	Higher value of transactions
Payment technology	↑	↓	Less need for money in transactions
Wealth	↑	↑	More resources to put into money
Riskiness of other assets	↑	↑	Money relatively less risky and so more desirable
Inflation risk	↑	↓	Money relatively more risky and so less desirable
Liquidity of other assets	↑	↓	Money relatively less liquid and so less desirable

Summary of the variables that shift money demand and the direction of their effects.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

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Empirical Evidence and Policy Implications

Why stability matters

- ▶ If money demand is stable:
 - ▶ Velocity is predictable
 - ▶ Central banks can target money supply
- ▶ If money demand is unstable:
 - ▶ Velocity fluctuates
 - ▶ Central banks focus on interest rates

Historical Evidence

- ▶ Before the 1970s:
 - ▶ Money demand relatively stable
 - ▶ Central banks targeted money supply
- ▶ After the 1970s:
 - ▶ Financial innovation destabilized money demand
 - ▶ Central banks switched to targeting interest rates

Chapter summary

- ▶ The equation of exchange $MV = PY$ becomes the quantity theory once velocity is constant and output is at its full-employment level: money growth drives the price level.
- ▶ In growth rates, $\pi = \% \Delta M - \% \Delta Y$; across countries and over long periods, inflation and money growth move together.
- ▶ Budget deficits cause inflation only when they are financed by printing money (ΔMB) rather than by selling bonds (ΔB).
- ▶ Keynes's liquidity preference theory gives real money demand $M^d/P = f(i, Y)$: negatively related to the interest rate, positively to income.
- ▶ Portfolio theories add wealth, risk, and liquidity of other assets; since the 1970s unstable money demand has pushed central banks toward interest-rate targets.